Vectors and Planes

Part 1/2: Coordinates, Lines and Plane Equations

CSC 1105 Mathematics for Computer Science

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Lecture Outline

- Points, position vectors, and displacement in three dimensions
- Cross products and their geometric interpretation
- Vector, parametric, and symmetric equations of lines
- Normal vectors and equations of planes
- Lines and planes in elementary computer graphics
- Revision questions

Module 5

Part	Focus
1/2	Three-dimensional coordinates; point and displacement vectors; cross products; equations of lines and planes; basic geometric tests.
2/2	Angles and distances involving lines and planes; parallelism and intersection; geometric applications and computational methods.

Learning Outcomes

By the end of this lecture, you should be able to:

1. Identify points, position vectors, and displacement vectors in $\mathbb{R}^3$.
2. Compute and interpret the cross product of two 3D vectors.
3. Construct and convert among standard equations of a line.
4. Determine an equation of a plane from geometric information.
5. Test whether a point lies on a line or a plane.
6. Apply line and plane representations to a simple ray–plane problem.

Connection to Module 4

From **Vectors and Vector Spaces**, recall:

- A vector has components, magnitude, and direction.
- Addition and scalar multiplication act componentwise.
- The dot product is $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$.
- $\mathbf{u} \cdot \mathbf{v} = 0$ for nonzero perpendicular vectors.

Note

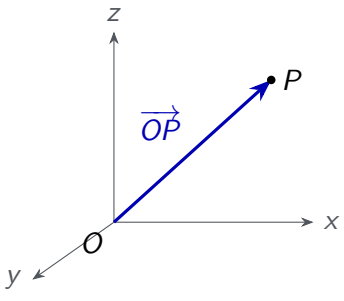
We now use vectors to describe **geometric objects**: points, straight lines, and flat planes in three dimensions.

Points and Position Vectors in $\mathbb{R}^3$

A point $P = (x, y, z)$ specifies a **location**. Its position vector from the origin is

$$\overrightarrow{OP} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

For example, $P = (2, 1, 3)$ has position vector $\overrightarrow{OP} = (2, 1, 3)^T$. **Distinction:** a point is a location; a vector is a displacement or direction.



Displacement Between Two Points

For $A = (x_1, y_1, z_1)$ and $B = (x_2, y_2, z_2)$,

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix}.$$

Example

Let $A = (1, 2, -1)$ and $B = (4, -2, 3)$. Then

$$\overrightarrow{AB} = \begin{bmatrix} 4 - 1 \\ -2 - 2 \\ 3 - (-1) \end{bmatrix} = \boxed{\begin{bmatrix} 3 \\ -4 \\ 4 \end{bmatrix}}.$$

The reverse displacement is $\overrightarrow{BA} = -\overrightarrow{AB}$.

Distance and Midpoint in Three Dimensions

Distance between points:

$$d(A,B) = \left\|\overrightarrow{AB}\right\| = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}.$$

Midpoint:

$$M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right).$$

Example

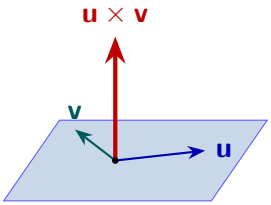
For $A=(1,0,2)$ and $B=(3,4,4)$: $\quad d(A,B) = \sqrt{4+16+4} = 2\sqrt{6}, \quad M=(2,2,3).$

Why Do We Need a Cross Product?

The geometric question

Given two directions in 3D, how can we find a vector perpendicular to *both*?

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$, the **cross product** $\mathbf{u} \times \mathbf{v}$ produces such a vector. It helps construct a **normal vector** to a plane.



Cross Product: Derivation Using Determinants

For $\mathbf{u} = (u_1, u_2, u_3)^T$ and $\mathbf{v} = (v_1, v_2, v_3)^T$:

1. **Express as a determinant:**

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

2. **Expand along the first row** (signs $+, -, +$):

$$\mathbf{u} \times \mathbf{v} = \mathbf{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \mathbf{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \mathbf{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

3. **Evaluate the 2×2 determinants:**

$$\begin{aligned} \mathbf{u} \times \mathbf{v} = &(u_2 v_3 - u_3 v_2)\mathbf{i} \\ &+ (u_3 v_1 - u_1 v_3)\mathbf{j} \\ &+ (u_1 v_2 - u_2 v_1)\mathbf{k} \end{aligned}$$

Cross Product: Formula

For $\mathbf{u} = (u_1, u_2, u_3)^T$ and $\mathbf{v} = (v_1, v_2, v_3)^T$,

$$\boxed{\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}}$$

An equivalent determinant mnemonic is

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Remember

Order matters: $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$.

Example: Cross Product

Let $\mathbf{u} = (1, 2, 0)^T$ and $\mathbf{v} = (0, 1, 3)^T$. Then

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2(3) - 0(1) \\ 0(0) - 1(3) \\ 1(1) - 2(0) \end{bmatrix} = \boxed{\begin{bmatrix} 6 \\ -3 \\ 1 \end{bmatrix}}.$$

Check perpendicularity:

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 1(6) + 2(-3) + 0(1) = 0,$$

$$\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0(6) + 1(-3) + 3(1) = 0.$$

Note

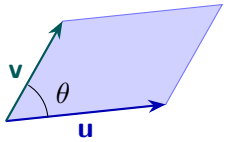
The cross product is perpendicular to each input vector.

Magnitude and Geometric Meaning

For θ the angle between nonzero vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\boxed{\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta.}$$

- Parallelogram area: $\|\mathbf{u} \times \mathbf{v}\|$.
- Triangle area: $\frac{1}{2} \|\mathbf{u} \times \mathbf{v}\|$.
- Parallel vectors have $\mathbf{u} \times \mathbf{v} = \mathbf{0}$.



Example: Area of a Triangle

Let $A = (0, 0, 0)$, $B = (2, 0, 0)$, and $C = (0, 3, 0)$.

$$\overrightarrow{AB} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \qquad \overrightarrow{AC} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix}.$$

Area

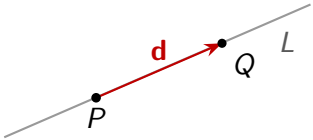
$$\text{Area of } \triangle ABC = \frac{1}{2} \left\| \overrightarrow{AB} \times \overrightarrow{AC} \right\| = \boxed{3 \text{ square units}}.$$

From a Direction to a Line

A straight line in $\mathbb{R}^3$ is determined by:

- a point P on the line, with position vector $\mathbf{p}$;
- a nonzero **direction vector d**.

Every point on the line is reached by moving some real multiple of $\mathbf{d}$ from P.



Vector Equation of a Line

Definition

Given point $\mathbf{p} \in \mathbb{R}^3$ and direction $\mathbf{d} \neq \mathbf{0}$, the line is

$$\boxed{\mathbf{r}(t) = \mathbf{p} + t\mathbf{d}, \quad t \in \mathbb{R}.}$$

Example:

$$\mathbf{r}(t) = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}.$$

At $t = 0$, we obtain $(1, -2, 3)$; at $t = 2$, we obtain $(5, 0, 1)$.

Parametric Equations of a Line

Expand the vector equation componentwise:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + t \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \Longleftrightarrow \quad \boxed{\begin{aligned} x &= x_0 + at, \\ y &= y_0 + bt, \\ z &= z_0 + ct. \end{aligned}}$$

Example

For $\mathbf{r}(t) = (1, -2, 3)^T + t(2, 1, -1)^T$,

$$x = 1 + 2t, \qquad y = -2 + t, \qquad z = 3 - t.$$

Symmetric Form of a Line

Eliminating t from $x = x_0 + at$, $y = y_0 + bt$, $z = z_0 + ct$ gives

$$\boxed{\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}} \qquad \text{when } a, b, c \neq 0.$$

Example:

$$\frac{x - 1}{2} = \frac{y + 2}{1} = \frac{z - 3}{-1}.$$

Zero direction components

Never divide by zero. If $\mathbf{d} = (2, 0, -1)^T$, use

$$\frac{x - 1}{2} = \frac{z - 3}{-1}, \qquad y = -2.$$

Line Through Two Distinct Points

Two distinct points A and B define a line with direction $\mathbf{d} = \overrightarrow{AB} = \mathbf{b} - \mathbf{a}$.

$$\boxed{\mathbf{r}(t) = \mathbf{a} + t(\mathbf{b} - \mathbf{a}), \quad t \in \mathbb{R}.}$$

Example

For $A = (1, 0, 2)$ and $B = (3, -1, 5)$,

$$\mathbf{d} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{r}(t) = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

At $t = 1$, the line passes through B.

Point-on-Line Test

Does $Q = (5, -2, 8)$ lie on the line through $A = (1, 0, 2)$ with direction $\mathbf{d} = (2, -1, 3)^T$?

$$\begin{bmatrix} 5 \\ -2 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \quad \Longrightarrow \quad \begin{cases} 5 = 1 + 2t, \\ -2 = -t, \\ 8 = 2 + 3t. \end{cases}$$

All three equations give $t = 2$.

Q lies on the line.

General test

A point lies on a parametric line only if **one common parameter value** satisfies every coordinate equation.

The Same Line Has Many Equations

Consider the line

$$\mathbf{r} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}.$$

Using a different point on it, $B = (3, -1, 5)$, and a rescaled direction gives

$$\mathbf{r} = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}.$$

- Both describe exactly the same geometric line.
- A nonzero scalar multiple of the direction is still a direction.
- A different point *on the line* is equally valid.

Note: A geometric line is unique; its vector representation is not.

Exercise 1: Lines

Task A. Find vector and parametric equations of the line through $A = (2, 1, -1)$ and $B = (4, 5, 2)$.

Task B. Does $Q = (6, 9, 5)$ lie on that line? Justify your answer with a common parameter.

Task C. Express

$$\mathbf{r}(t) = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$$

in symmetric form, without dividing by zero.

Exercise 1: Answers

A. The direction is $B - A = (2, 4, 3)^T$:

$$\mathbf{r}(t) = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}.$$

B. $Q - A = (4, 8, 6)^T = 2(2, 4, 3)^T$: $\boxed{t = 2}$, so Q lies on the line.

C. $x = 3t$, $y = 1$, $z = 2 - 2t$. Hence

$$\boxed{\frac{x}{3} = \frac{z - 2}{-2}, \qquad y = 1.}$$

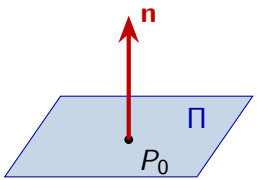
What Is a Plane?

A **plane** is a flat, two-dimensional geometric surface extending indefinitely in $\mathbb{R}^3$.

A plane can be specified by

- a point P_0 lying on it, and
- a nonzero vector **n** perpendicular to it.

Such a vector is called a **normal vector**.



Point–Normal Equation of a Plane

Let $P_0 = (x_0, y_0, z_0)$ be on the plane and let its normal be $\mathbf{n} = (a, b, c)^T \neq \mathbf{0}$. Every point $P = (x, y, z)$ on the plane satisfies

$$\overrightarrow{P_0P} \cdot \mathbf{n} = 0.$$

Therefore,

$$\boxed{a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.}$$

Note

A plane consists of all points whose displacement from P_0 is perpendicular to its normal.

Cartesian Equation of a Plane

Expanding point–normal form yields

$$\boxed{ax + by + cz = d, \qquad (a, b, c) \neq (0, 0, 0).}$$

- The coefficient vector $\mathbf{n} = (a, b, c)^T$ is normal to the plane.
- d determines the plane's position relative to the origin.
- $d = 0$ means that the plane passes through the origin.

Example

The plane $2x - y + 3z = 7$ has normal vector $\mathbf{n} = (2, -1, 3)^T$.

Example: Plane Through a Point

Find the plane through $P_0 = (1, 2, -1)$ with normal $\mathbf{n} = (2, -1, 3)^T$.

Step 1: Write the point–normal equation:

$$2(x - 1) - (y - 2) + 3(z + 1) = 0.$$

Step 2: Expand and simplify:

$$2x - 2 - y + 2 + 3z + 3 = 0.$$

$$\boxed{2x - y + 3z = -3.}$$

Check: at $(1, 2, -1)$, $2 - 2 - 3 = -3$.

Point-on-Plane Test

Consider the plane $2x - y + 3z = -3$.

Point $Q = (0, 3, 0)$

Substitute: $2(0) - 3 + 3(0) = -3$.
Therefore, Q lies on the plane.

Point $R = (0, 0, 0)$

Substitute: $2(0) - 0 + 3(0) = 0 \neq -3$.
Therefore, R is not on the plane.

Method: a point belongs to a plane exactly when its coordinates satisfy the plane equation.

Plane Through Three Non-collinear Points

Given three points A, B, C not on one straight line:

1. Form two in-plane direction vectors: $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AC}$.
2. Compute $\mathbf{n} = \mathbf{u} \times \mathbf{v}$.
3. Use A with the normal in the point–normal equation.

$$\boxed{\big((\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})\big) \cdot (\mathbf{r} - \mathbf{a}) = 0.}$$

Condition

The points must be non-collinear; otherwise the cross product is zero and no unique plane is determined.

Example: Plane Through Three Points

Let $A = (1, 0, 0)$, $B = (0, 1, 0)$, and $C = (0, 0, 1)$.

$$\overrightarrow{AB} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \qquad \overrightarrow{AC} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Their cross product is

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Through $A = (1, 0, 0)$, the equation is

$$(x - 1) + y + z = 0 \quad \Longrightarrow \quad \boxed{x + y + z = 1.}$$

All three given points satisfy this equation.

Parametric Form of a Plane

A plane may also be described by a point $\mathbf{p}$ and two **linearly independent** directions $\mathbf{u}, \mathbf{v}$:

$$\boxed{\mathbf{r}(s,t) = \mathbf{p} + s\mathbf{u} + t\mathbf{v}, \qquad s,t \in \mathbb{R}.}$$

Example

The plane $z = 2$ has parametric equation

$$\mathbf{r}(s,t) = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} + s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

Here $x = s$, $y = t$, and $z = 2$.

Connecting Parametric and Cartesian Forms

Suppose

$$\mathbf{r}(s,t) = \begin{bmatrix}1\\0\\0\end{bmatrix} + s\begin{bmatrix}-1\\1\\0\end{bmatrix} + t\begin{bmatrix}-1\\0\\1\end{bmatrix}.$$

The directions are $\mathbf{u} = (-1,1,0)^T$ and $\mathbf{v} = (-1,0,1)^T$.

$$\mathbf{n} = \mathbf{u} \times \mathbf{v} = \begin{bmatrix}1\\1\\1\end{bmatrix}.$$

Using point $(1,0,0)$ yields

$$\boxed{x + y + z = 1.}$$

Note: Two in-plane directions determine a normal using their cross product.

Demonstration: A Line Meets a Plane

Consider the line

$$\mathbf{r}(t) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

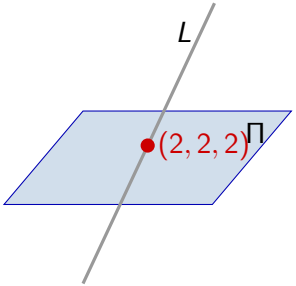
and the plane $x + y + z = 6$. Substitute $x = y = z = t$:

$$t + t + t = 6 \implies t = 2.$$

Thus, the line intersects the plane at

$$\boxed{(2, 2, 2)}.$$

A systematic classification of line–plane relationships is covered in Part 2.



Computer Science Application: Ray–Plane Intersection

In 3D graphics, a **ray** is a half-line from a starting point:

$$\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}, \qquad t \geq 0.$$

For a plane with normal $\mathbf{n}$ through $\mathbf{p}$, substitute the ray into $\mathbf{n} \cdot (\mathbf{r} - \mathbf{p}) = 0$:

$$\mathbf{n} \cdot (\mathbf{o} + t\mathbf{d} - \mathbf{p}) = 0 \quad \Longrightarrow \quad \boxed{t = \frac{\mathbf{n} \cdot (\mathbf{p} - \mathbf{o})}{\mathbf{n} \cdot \mathbf{d}}.}$$

This formula applies when $\mathbf{n} \cdot \mathbf{d} \neq 0$; the intersection lies on the ray only when $t \geq 0$.

Ray–Plane Intersection: Numerical Example

Ray origin $\mathbf{o} = (0,0,0)^T$, direction $\mathbf{d} = (1,1,1)^T$. The plane $z = 5$ has $\mathbf{p} = (0,0,5)^T$, $\mathbf{n} = (0,0,1)^T$.

$$t = \frac{\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}}{\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}} = \frac{5}{1} = 5.$$

The ray reaches the plane at

$$\mathbf{r}(5) = (0,0,0)^T + 5(1,1,1)^T = \boxed{(5,5,5)^T}.$$

Interpretation: the point can be used to determine where a viewing ray meets a planar surface.

Python Example: Checking a Point on a Plane

For plane $2x - y + 3z = -3$, let $\mathbf{n} = (2, -1, 3)^T$.

NumPy computation

```
import numpy as np

n = np.array([2.0, -1.0, 3.0])
point = np.array([0.0, 3.0, 0.0])
d = -3.0

is_on_plane = np.isclose(n @ point, d)
print(is_on_plane) # True
```

`n @ point` computes the dot product.
Use `isclose` rather than exact equality with floating-point values.

Exercise 2: Planes

Task A. Determine the plane through $P = (2, -1, 0)$ with normal $\mathbf{n} = (1, 3, -2)^T$.

Task B. Test whether $Q = (1, 0, 1)$ lies on that plane.

Task C. Find the equation of the plane through $A = (0, 0, 1)$, $B = (1, 0, 1)$ and $C = (0, 2, 1)$.

Task D. Write a parametric equation for the plane found in Task C.

Exercise 2: Answers

A. Point–normal form:

$$(x-2)+3(y+1)-2z=0 \implies \boxed{x+3y-2z=-1}.$$

B. At $Q=(1,0,1)$, $1+0-2=-1$: **yes**.

C. $\overrightarrow{AB}=(1,0,0)^T$, $\overrightarrow{AC}=(0,2,0)^T$, so a normal is $(0,0,2)^T$ and the plane is $\boxed{z=1}$.

D. $\boxed{\mathbf{r}(s,t)=\begin{bmatrix}0\\0\\1\end{bmatrix}+s\begin{bmatrix}1\\0\\0\end{bmatrix}+t\begin{bmatrix}0\\2\\0\end{bmatrix}.}$

Lecture Summary

- A position vector describes a point relative to the origin.
- The cross product yields a vector perpendicular to two 3D directions.
- A line is represented by $\mathbf{r} = \mathbf{p} + t\mathbf{d}$.
- A plane is represented by $\mathbf{n} \cdot (\mathbf{r} - \mathbf{p}) = 0$.
- Two independent in-plane directions produce a normal via their cross product.
- These geometric equations support basic graphics calculations.

Questions?